

Fig. 2: Solid-state relay module and its connections

This means that, when you read the input value using `GPIO.input`, 'False' will be returned if the button is pressed. Each GPIO pin has software configurable pull-up and pull-down resistors.

The `event_detected()` function is designed to be used in a loop with other things, but unlike polling, it is not going to miss the state change of an input while the CPU is busy working on other things. This could be useful in something like a quiz game where there is a main loop listening and responding to GUI events on a timely basis. For example, falling-edge detection on a channel can be written as:

```
GPIO.add_event_detect(channel, GPIO.FALLING)
do_something()
if GPIO.event_detected(channel):
    do_something()
```

`GPIO.remove_event_detect` channel is used to latch an input push-button once it is pressed. Various other commands and details can be found from the link <http://sourceforge.net/p/raspberry-gpio-python/wiki/Inputs/>

A solid-state relay (SSR) is an electronic switching device that switches on or switches off when a small external voltage is applied across its control terminals. An SSR has a sensor

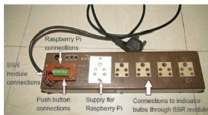


Fig. 3: System's various connecting points



Fig. 4: Author's prototype

that responds to an appropriate input (control signal), a solid-state electronic switching device that switches power to the load circuitry and a coupling mechanism to enable the control signal to activate this switch without mechanical parts.

The relay may be designed to switch either AC or DC to the load. It serves the same function as an electromechanical relay, but has no moving parts. Solid-state relay details can be found from the link https://en.wikipedia.org/wiki/Solid-state_relay.

The code `rasquiz.py` written in Python language detects the push-button switches that are connected to the glowing bulbs, which show the fastest finger first contestants' sequence, and to reset the system.

Construction and testing

Each bulb holder and push-button switch is mounted on a suitable box,

with separate pair of wires for the bulb holder and the switch. GPIO board numbering is used throughout the project. Push-button switches' wires are connected to GPIO pins 11, 13, 15, 16 and common to ground pin 9 of Raspberry Pi as shown in Fig. 1.

1. A 5V, 4-channel solid-state relay module (refer Fig. 2), purchased from ebay.in, is connected to Raspberry Pi GPIO pins 18, 19, 21 and 23.

A 5V supply is given to relay module from Raspberry Pi. Coloured indicator bulbs (Bulb 1 through Bulb 4) are connected to the output of the relay board with 230V supply as shown in Fig. 1. System's various connecting points are shown in Fig. 3.

Run the `rasquiz.py` Python script on the terminal by giving the command:

```
$ sudo python rasquiz.py
```

The program executes from main subroutine. It prompts for user inputs like names of the teams and allowable time (in seconds) to respond to a question. Then it clears the screen and prompts you to continue or to stop. To continue the game, type 1 followed by pressing 'Enter' key. Raspberry Pi initialises the push-button switches' pins as input and clears the solid-state relay pins to switch off the indicators.

As soon as contestants press their switches, names of the respective teams with their rank numbers get displayed on LCD screen, along with glowing of respective indicator bulbs. If no contestant presses the switch within specified time, all the switches' pins get latched, so no more inputs will get registered and 'Time Up' message is displayed on screen.

By typing Ctrl + C keys at any time, the host or operator can stop the quiz without restarting the whole system or waiting till time up. When prompted, enter 1 to continue or 0 to quit the game. **BY**



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EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com